TATA MOTORS LIMITED

Company Research Report | Diploma Internship

Automotive Products | History | R&D | Manufacturing

TATA MOTORS

India's Automotive Leader

DIPLOMA FINAL-YEAR INTERNSHIP
COMPANY RESEARCH REPORT

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Internship organisation: Tata Motors Limited
Academic year: 2026–2027
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Submitted in partial fulfilment of the requirements for the Diploma final-year internship.

1. Introduction

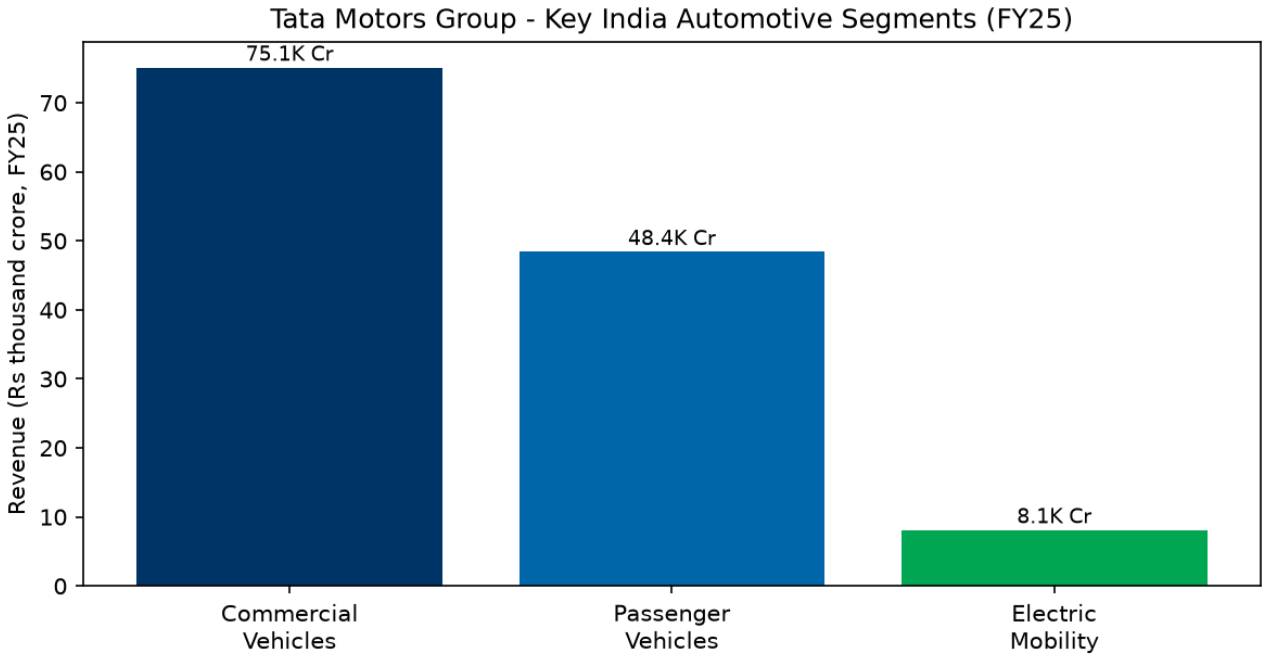
The automobile industry is one of the most important sectors of the Indian economy. It contributes to industrial development, employment, innovation, infrastructure, and trade. Tata Motors is a major manufacturer in passenger vehicles, commercial vehicles, electric mobility, and engineering. Established in 1945 as TELCO, it expanded from locomotives to cars, SUVs, trucks, buses, and EVs. This report studies background, segments, products, manufacturing, R&D, and challenges.

2. About Tata Motors

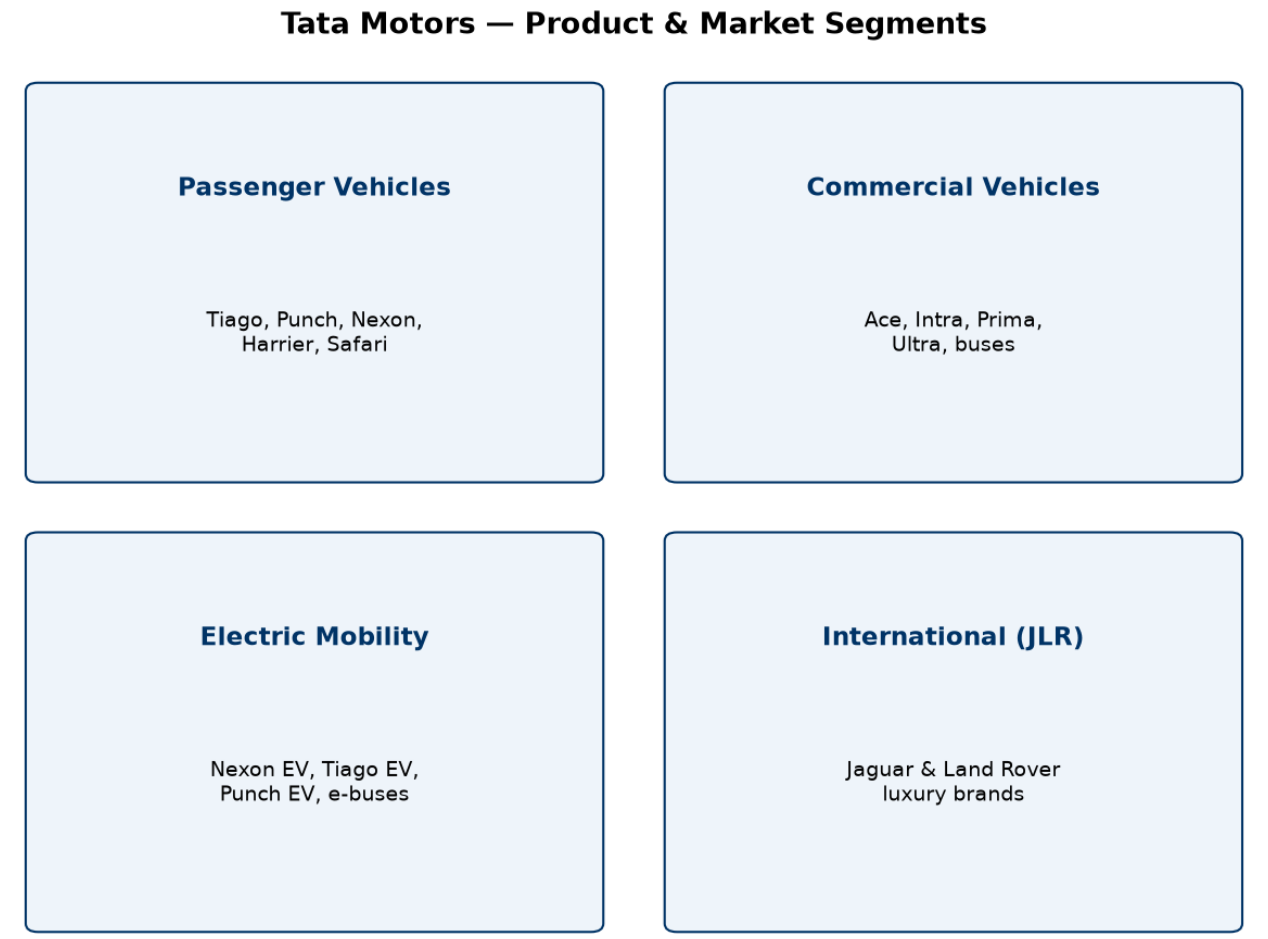
Tata Motors is part of the Tata Group. It designs, develops, manufactures, markets, and services vehicles for individuals, fleets, logistics, and public transport. It invests in R&D, electrification, connected technology, safety, and advanced manufacturing.

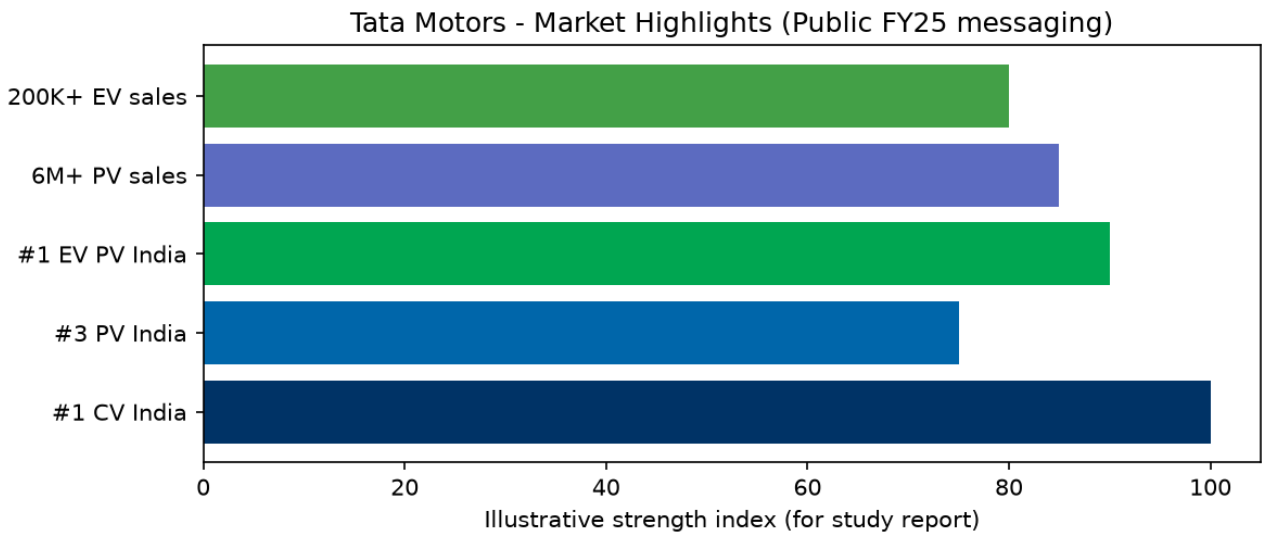
3. Company Overview

Business segments (FY25): Tata Commercial Vehicles, Tata Passenger Vehicles, electric mobility, and Jaguar Land Rover. From October 2025, CV and PV were demerged into focused listed companies (NCLT-approved). CV includes trucks, buses, Ace/Intra; PV includes Tiago, Nexon, Harrier; EV includes Nexon EV and e-buses; JLR is the global luxury business.



Source: Tata Motors FY25 investor materials. JLR reported as separate global segment.





4. History and Development of Tata Motors

Covers 1945 TELCO foundation, 1954 CV entry, 1991 Sierra, 1998 Indica, 2004 Daewoo, 2005 Ace, 2008 JLR, 2016 Tiago/IMPACT, 2019 Nexon EV/ZIPTRON, 2023 Sanand capacity, 2024 acti.ev, 2025 Curvv and business demerger. See detailed timeline figures.

Milestones in Tata Motors History



Detailed History & Development of Tata Motors

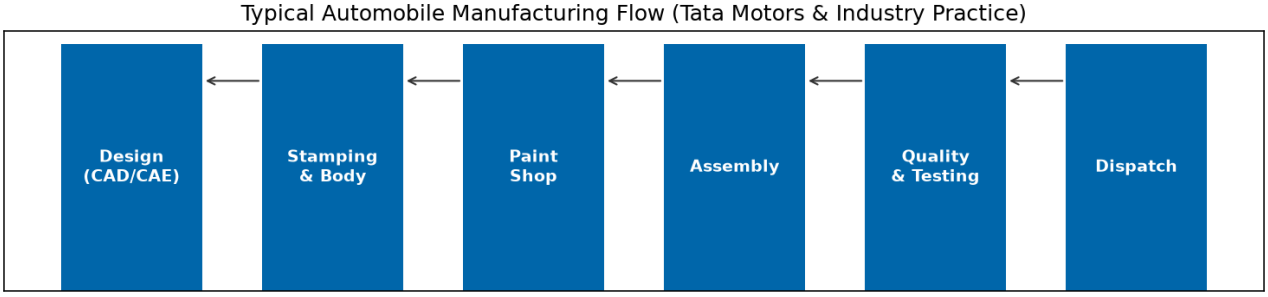
- **1945** — TELCO incorporated; locomotive & engineering
- **1954** — Commercial vehicles with Daimler-Benz
- **1986** — Iconic Tata 407 LCV
- **1991** — Tata Sierra - entry into passenger vehicles
- **1998** — Tata Indica - first fully indigenous Indian car
- **2004** — Daewoo Commercial Vehicles acquisition (Korea)
- **2005** — Tata Ace - India's first mini truck
- **2008** — Jaguar Land Rover acquired from Ford
- **2016** — Tiago & IMPACT design philosophy
- **2019** — Nexon EV with ZIPTRON technology
- **2020** — BS VI transition; Safari relaunch
- **2023** — Ford Sanand plant acquisition; capacity expansion
- **2024** — acti.ev platform; 6M+ cumulative PV sales milestone
- **2025** — Curvv, Nexon iCNG; CV-PV demerger; Tamil Nadu plant

5. Manufacturing Methodology Used by Tata Motors

Lean manufacturing, Just-in-Time, Kaizen, TQM, automation and robotics, CAD/CAM, supply chain

management, environmental and safety management, EV high-voltage processes, and end-of-line quality testing including electrical and road tests.

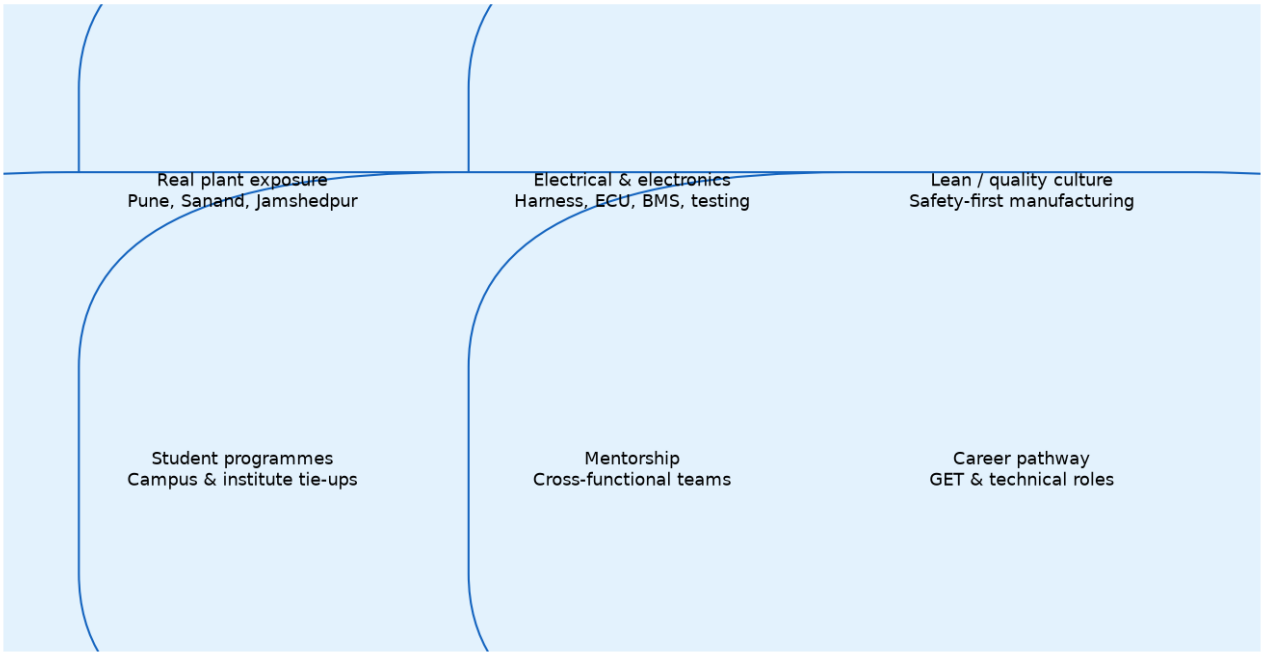
Typical Automobile Manufacturing Flow (Tata Motors & Industry Practice)



Executive Summary / Internship

Tata Motors is a leading choice for diploma internships in ECE: plant exposure, harness/ECU work, safety culture, and student programmes (joinus.tatamotors.com). FY25 milestones: 6M PV sales, 200K EV sales.

Why Internship at Tata Motors? (ECE / Diploma Students)



6. Products Manufactured by Tata Motors

Passenger: Tiago, Altroz, Tigor, Punch, Nexon, Harrier, Safari, Curvv. Commercial: Ace, Intra, Yodha, Ultra, Signa, Prima, Starbus. Electric: Nexon EV, Tiago EV, Punch EV, Ace EV, electric buses. Products serve personal mobility, goods transport, and public transport.

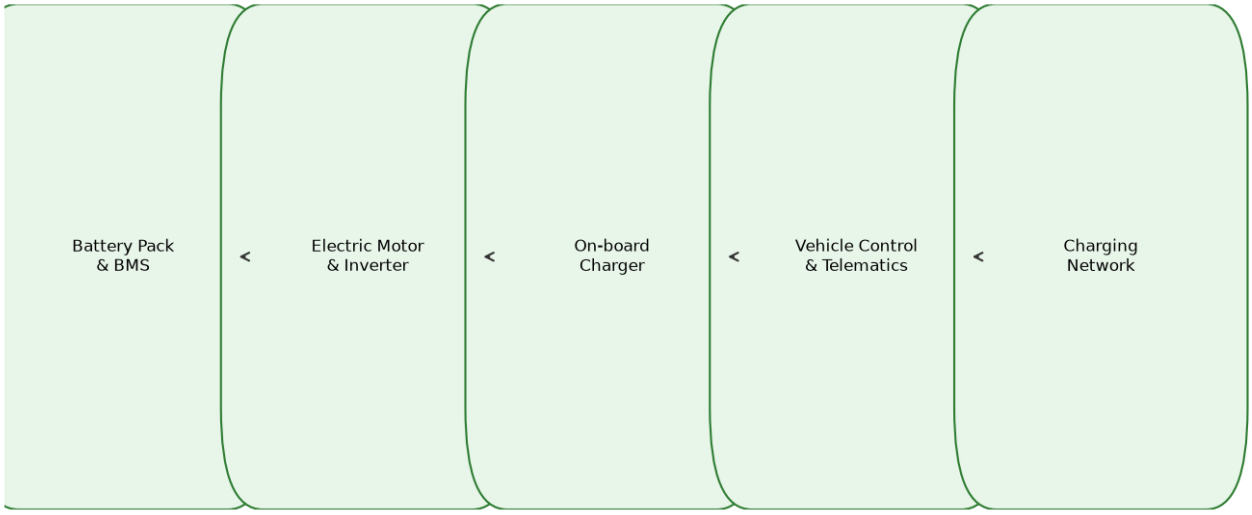
Key Products Manufactured by Tata Motors (Representative Portfolio)



Representative Tata Motors Product Families



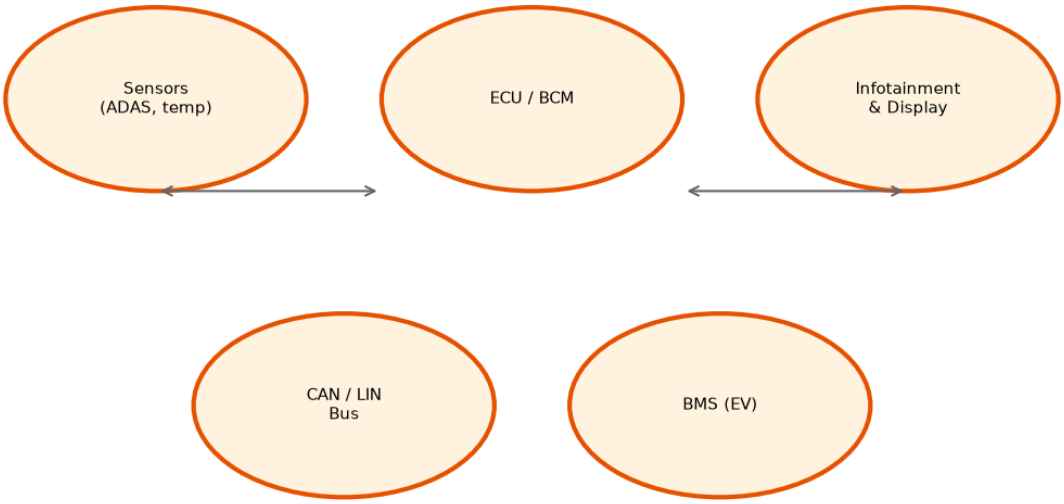
Electric Vehicle Value Chain (Tata Motors Focus Areas)



6.1 R&D and Electronics (ECE)

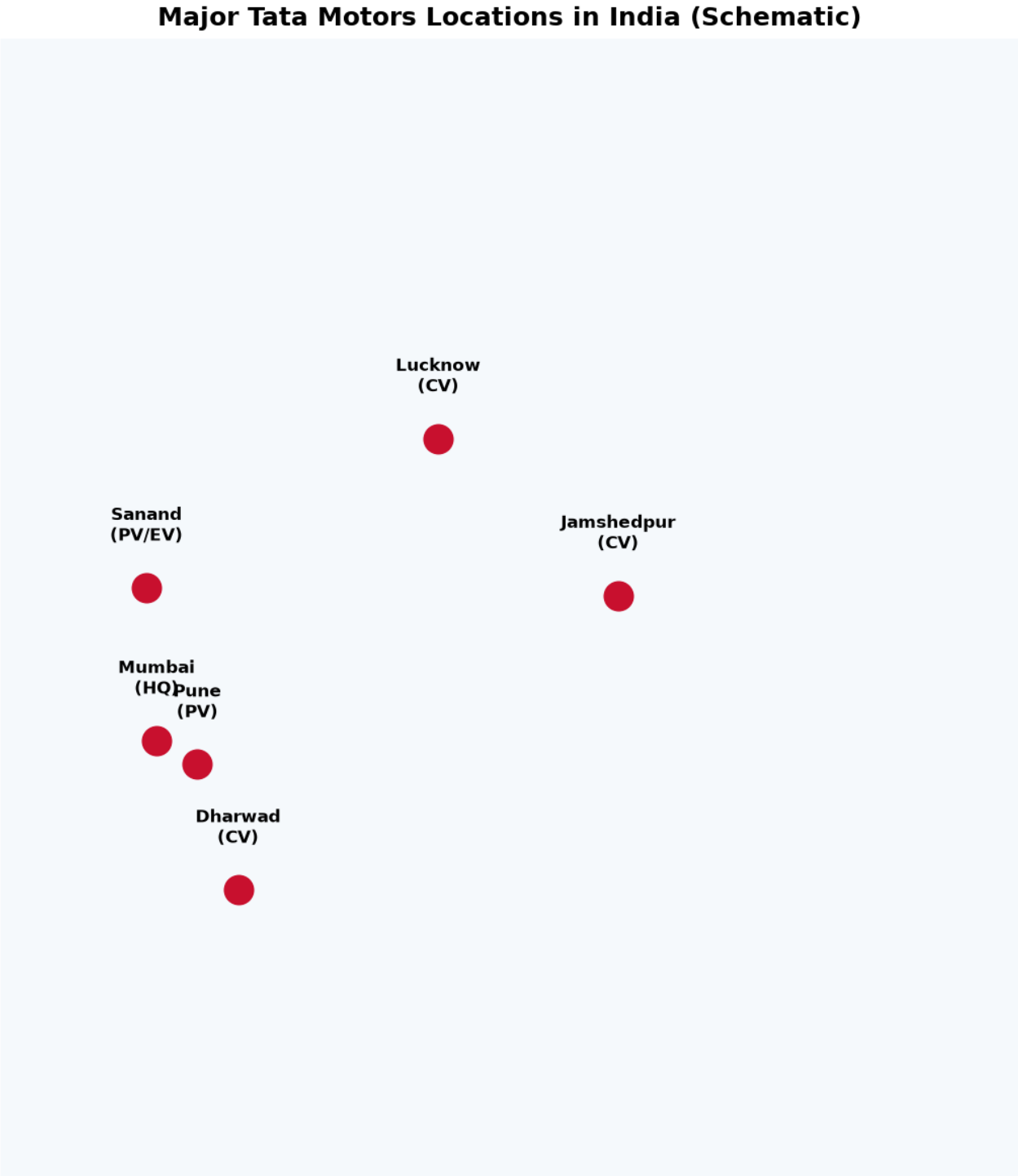
Modern vehicles use ECUs, BCM, BMS (EV), sensors, infotainment, telematics, CAN/LIN buses, and OTA updates. ECE students study embedded systems, power electronics, and production testing.

Electronics in Modern Tata Vehicles (ECE Relevance)



6.2 Plants in India

Major plants: Pune (PV), Jamshedpur and Lucknow (CV), Sanand (PV/EV), Dharwad (CV). Processes include stamping, body shop, paint, assembly, and quality gates.



7. Challenges and Future Opportunities

Challenges: competition, pricing, supply chain, quality, electrification, regulations. Opportunities: R&D, digitalisation, energy efficiency, battery recycling, connected vehicles.

8. Research Methodology

Descriptive document-based research using annual reports, investor presentations, product catalogues, and authorised internship observations. Limitations: confidential plant data is not publicly available.

9. Conclusion

Tata Motors is a diversified automotive leader in India with strong CV and PV portfolios, EV leadership,

and global presence through JLR. Its manufacturing and electronics integration offer valuable learning for diploma internship and industrial visits.

10. References

[1] Tata Motors Integrated Annual Report 2024-25.
[2] Tata Motors Q4 FY25 investor presentation.
[3] www.tatamotors.com
[4] Government of India automotive sector publications.